Learning Nonlinear Optical Dysthe Dynamics

Abstract

This outline concerns prediction across initial fields for optical Dysthe. Reference migration, learning methods and experiments remain to be completed. No new performance or novelty claim is made.

1 Question and scope

State one testable question about accuracy, cost, or a resolved wave mechanism. Identify the closest prior work and the proposed distinction. Keep optical Dysthe separate from water-wave and kinetic-plasma models.

2 Equation and reference problem

Use `optical-dysthe-127-v1` from the shared core. Record the equation, signs, coordinates, coefficients, initial-field family and domain. Report grid, time-step and domain checks; state the limits of a periodic box as an approximation to whole space.

3 Learning method and protocol

Describe the selected method and baselines. Declare prediction inputs, normalization, loss, optimizer, early stopping, checkpoints, seeds and hardware. Keep each initial condition and its refinements in one split. Select models using validation; reserve a locked test set for final claims.

4 Results to establish

Provide convergence, held-out field and phase errors, growth or event timing, physical diagnostics, rollout error and cost comparisons. Report failures and variability across cases and seeds. Conservation does not establish phase accuracy. Separate training and inference costs.

5 Interpretation and reproducibility

Connect each claim to its run and figure. Distinguish computational improvement from wave-physics discovery. Do not present trends as theorems or infer long-time laws from unresolved runs. Freeze code revisions; record configurations, data and checkpoint hashes, artifact locations and software versions. Use PDF figures and retain compact numerical figure data. The claim ledger is initially empty.